Technical Report: Periodic-1D Rank-Two Band Equivalence versus Hopping Locality

Project Koios

Development migration draft; independently verified unsealed calculation discussed explicitly

Abstract

This report gives a deliberately exhaustive mathematical account of the finite, controlled periodic-one-dimensional rank-two study. It defines the ambient parent Hamiltonian, retained spectral subspace, periodic frame construction, represented Hamiltonian, gauge action, restricted comparison problem, Fourier-locality measures, momentum and amplitude refinement rules, ordinary-lattice truncation experiment, closed failure outcomes, canonical aggregate record, and independent-verification boundary. It records both the failed first verification attempt and its preserved artifacts, then the distinct authorized no-overwrite successor execution. The successor passed stored independent verification at the unchanged installed tolerance and produced the authenticated six-figure inventory shown here. This establishes bounded finite software-route agreement and artifact integrity, not scientific validation, evidence sealing, or publication acceptance.

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1 Scope, questions, and evidence boundary

1.1 Scientific question

The study asks a representation-sensitive question inside one fixed rank-two physical subspace. Suppose two matrix fields are related pointwise by a unitary change of retained basis and therefore have identical ordered eigenvalues at every crystal momentum. Must they have the same locality when each is Fourier transformed in its own declared retained frame? The answer is no: pointwise spectral equivalence constrains eigenvalues but not the momentum dependence of the representing basis. A smooth, momentum-dependent gauge action can preserve every band energy while redistributing matrix-valued hopping power over cell displacements.

The numerical study makes that statement quantitative in four separate channels:

1. a bounded parent-hopping family, which changes the ambient parent dispersion while preserving all other controls;
2. nested momentum-mesh refinement, which measures finite-mesh stability and residual consistency without changing the physical model;
3. deterministic adaptive amplitude refinement, which measures interpolation defects in the gauge-attack parameter and requires disjoint validation before convergence; and
4. ordinary-lattice truncation, which measures omitted hopping power, represented Hamiltonian error, and sorted-band error on a mesh disjoint from the construction mesh.

These channels are retained together but are not collapsed into one error number. Parent-model variation, reciprocal discretization, parameter interpolation, and model reduction answer different questions.

1.2 What "band equivalent" means here

For the reference retained Hamiltonian $H(q)$ and comparison Hamiltonian $A_a(q)$, band equivalence means

$$\operatorname{spec} A_a(q) = \operatorname{spec} H(q) \quad \text{for every evaluated } q, \tag{1}$$

with eigenvalues compared in sorted order. It does not mean $A_a(q) = H(q)$ as matrices, it does not identify their Fourier coefficients, and it does not permit an arbitrary momentum-dependent alignment during the comparison. The scientific comparison is intentionally restricted to one momentum-independent real rotation, with a separately optimized nonnegative scalar where that scalar is part of the declared question.

This restriction is substantive. If the same momentum-dependent gauge transformation used to construct A_a were admitted during alignment, the comparison would be trivial: the fields could be returned to pointwise equality by construction. The restricted problem instead asks what can be represented by one global rank-two coordinate choice.

1.3 Finite-control status

Everything computed in this study is finite. The parent is a synthetic 4×4 Hermitian matrix field. Its retained subspace has rank two. Construction uses a finite centered uniform momentum mesh. Fourier coefficients form a finite cyclic inventory before an explicit even-mesh cyclic-to-ordinary conversion. Amplitude studies use a finite interval $0 \le a \le \pi/4$, and truncation studies use finitely

many integer cutoffs. No thermodynamic limit, continuum limit, complete extremum certificate, or material calibration follows merely from these finite operations.

Two unsealed execution directories are presently retained. The first preserves a failed independent-verification attempt and has no figure inventory. The authorized no-overwrite successor preserves the same canonical calculation bytes, passed aggregate independent verification at the unchanged installed tolerance, and produced the closed six-figure inventory through stored admission. Neither directory is a reviewed release, a sealed evidence package, a publication result, or a material prediction. Section 5 keeps the failed verification history, the successful bounded route agreement, and the remaining unresolved scientific channels distinct.

1.4 Integrity, verification, and scientific validity

Three evidentiary notions are kept distinct:

Canonical integrity. Canonical typed JSON and SHA-256 identify exact bytes and reject alteration or noncanonical re-encoding. A digest says nothing by itself about whether a formula is physically appropriate.

Independent route agreement. A verifier reconstructs the same finite observables through direct formulas that avoid the production scientific owners. Passing would establish bounded numerical agreement under the installed comparison policy.

Scientific validity. Applicability of the model, adequacy of the controls, convergence of relevant limits, interpretation of observables, and suitability for a scientific claim require arguments beyond byte identity and software-route agreement.

The first candidate satisfies neither a passing aggregate-verification claim nor a rendered-artifact claim; its failure remains a diagnostic fact. The successor satisfies the bounded independent-route and authenticated-rendering contracts, but that software agreement does not upgrade the finite calculation to scientific validation or sealed evidence.

1.5 Ownership boundary

Quantum ESPRESSO and Wannier90 are not used by this synthetic study. The study does not replace either package. Project Koios Optimization supplies unit-free numerical algorithms, PhysKit supplies physical and finite-periodic semantics, and `ksdft2effmass` owns the installed model, policy thresholds, closed outcome variants, aggregate retention, verification admission, and calculation-facing visualization semantics. This division matters because independent verification must not call the same production owner whose result it is meant to check.

2 Model, state spaces, and conventions

2.1 Reduced momentum and physical wavevector

The Brillouin coordinate used by every installed numerical owner is the dimensionless reduced crystal momentum

$$q \in \mathcal{B} = [-\tfrac{1}{2}, \tfrac{1}{2}), \qquad q \sim q + 1. \tag{2}$$

If a_0 is a physical lattice period, $G = 2\pi/a_0$ is the reciprocal-lattice period and k is the inverse-length wavevector, then

$$q = \frac{k}{G} = \frac{ka_0}{2\pi}, \qquad k = qG. \tag{3}$$

Accordingly, all band functions in this report are written as $\lambda_n(q)$. The symbol k is reserved for a physical wavevector and is never used as a finite-mesh index. Integer mesh indices are denoted j, amplitude indices by r, and cell displacements by R.

For even N, the centered half-open construction mesh is

$$\mathcal{Q}_N = \left\{ q_j = -\frac{1}{2} + \frac{j}{N} : j = 0, \ldots, N-1 \right\}. \tag{4}$$

The half-open convention avoids duplicating the periodically equivalent endpoints. All interval statements involving q are understood modulo one.

2.2 Ambient and retained state spaces

The ambient cell space is $\mathbb{C}^4$ in a fixed ordered four-orbital basis. The retained fiber at momentum q is a two-dimensional spectral subspace $\mathcal{S}(q) \subset \mathbb{C}^4$. A frame $F(q) \in \mathbb{C}^{4\times 2}$ is an isometric coordinate map from the retained coordinate space $\mathbb{C}^2$ into $\mathcal{S}(q)$:

$$F(q)^\dagger F(q) = I_2, \qquad P(q) = F(q)F(q)^\dagger. \tag{5}$$

The physical subspace is represented by the projector $P(q)$ and is unchanged by a right unitary action $F(q) \mapsto F(q)W(q)$. The represented 2×2 Hamiltonian, however, changes by conjugation. This distinction between subspace and representation is the central conceptual layer of the study.

2.3 Finite-range four-orbital parent

Two continuous maps $f_0, f_1 : X \to Y$ are *homotopic* when there exists a continuous map

$$\mathcal{H} : X \times [0,1] \longrightarrow Y \quad \text{such that} \quad \mathcal{H}(x,0) = f_0(x), \qquad \mathcal{H}(x,1) = f_1(x). \tag{6}$$

The map $\mathcal{H}$ is called a *homotopy* between f_0 and f_1. In this study, $X = \mathcal{B}$ is the period-one Brillouin circle, $Y = \mathrm{Herm}(4)$ is the space of 4×4 Hermitian matrices, and the interpolation parameter is $\rho \in [0,1]$. Consequently, if $(q,\rho) \mapsto K_\rho(q)$ is continuous, then the endpoint parent fields K_0 and K_1 are homotopic as Hermitian matrix-valued maps. This use of *homotopy* means a continuous parent-model interpolation only. It does not by itself establish a gap-preserving homotopy or topological equivalence: that would require the relevant spectral gap to remain open for every continuous pair (q,ρ), whereas the calculation applies its external-gap gate only on finite sampled inventories.

Let $z(q) = e^{2\pi i q}$. The installed parent-hopping homotopy is the continuous map

$$K_\rho(q) = \begin{pmatrix} L_0 + \rho[z(q)L_1 + \overline{z(q)}L_1^\dagger] & \rho z(q) C \\ \rho\overline{z(q)}C^\dagger & U_0 + \rho[z(q)U_1 + \overline{z(q)}U_1^\dagger] \end{pmatrix}, \qquad 0 \le \rho \le 1, \tag{7}$$

where

$$L_0 = \begin{pmatrix} -1.20 & 0.06 \\ 0.06 & -0.35 \end{pmatrix}, \qquad L_1 = \begin{pmatrix} 0.15 & 0 \\ 0 & -0.10 \end{pmatrix}, \tag{8}$$

$$U_0 = \begin{pmatrix} 1.00 & 0.04 \\ 0.04 & 1.75 \end{pmatrix}, \qquad U_1 = \begin{pmatrix} 0.05 & 0 \\ 0 & -0.04 \end{pmatrix}, \tag{9}$$

$$C = \begin{pmatrix} 0.12 & 0 \\ 0 & 0.10 \end{pmatrix}. \tag{10}$$

All numerical entries use one arbitrary positive synthetic parent-energy unit. There is no conversion to electronvolts and no material parametrization. Equation (7) is exactly Hermitian because its lower-left block is the adjoint of its upper-right block and the diagonal blocks include conjugate hopping terms.

Equivalently,

$$K_\rho(q) = \sum_{R=-1}^{1} e^{2\pi i q R} T_R(\rho), \qquad T_{-R}(\rho) = T_R(\rho)^\dagger. \tag{11}$$

At $\rho = 0$ the ambient parent is onsite and independent of q. At $\rho = 1$ it is the full installed finite-range parent. Intermediate ρ values vary only the non-onsite parent hopping; the onsite blocks, state spaces, gates, tolerances, and gauge-amplitude inventory remain unchanged.

2.4 Spectral retention and external-gap gate

Let

$$\lambda_1(q) \le \lambda_2(q) \le \lambda_3(q) \le \lambda_4(q) \tag{12}$$

be the ordered eigenvalues of $K_\rho(q)$. The raw retained frame V_j consists of the first two normalized eigenvectors returned at q_j. The sampled external isolation gap is

$$\Delta_{\mathrm{ext},N}(\rho) = \min_{q_j \in \mathcal{Q}_N} \left[\lambda_3(q_j) - \lambda_2(q_j)\right]. \tag{13}$$

Retention is allowed to proceed only if $\Delta_{\mathrm{ext},N}(\rho) \ge \delta_{\mathrm{ext}}$. Failure of this gate is a scientific qualification stop, not a numerical transport failure and not a failed comparison threshold.

The raw eigenvectors are not presumed to define a smooth periodic gauge. Arbitrary eigenvector phases and rotations inside the rank-two eigenspace must be resolved before Fourier locality is meaningful. Section 3.2 specifies the closed-loop polar transport used for that purpose.

2.5 Represented retained Hamiltonian

Once a periodic transported frame $F_j = F(q_j)$ is available, the retained Hamiltonian is

$$H_j = F_j^\dagger K_\rho(q_j) F_j \in \mathbb{C}^{2\times 2}. \tag{14}$$

Its scalar and traceless parts are

$$m_j = \frac{1}{2}\operatorname{tr} H_j, \qquad X_j = H_j - m_j I_2, \qquad \operatorname{tr} X_j = 0. \tag{15}$$

The installed splitting scale is the discrete normalized traceless power

$$\kappa_N = \frac{1}{2N} \sum_{j=0}^{N-1} \|X_j\|_F^2 > 0. \tag{16}$$

Division by two is division by retained rank. Thus every squared Frobenius quantity in the main comparison has the dimensions of energy squared per retained state.

2.6 Gauge attack and exact pointwise band equivalence

The declared smooth gauge angle and real rank-two rotation are

$$\theta_a(q) = a\sin(2\pi q), \qquad \mathcal{R}(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad 0 \le a \le \frac{\pi}{4}. \tag{17}$$

The comparison field is

$$A_a(q) = \mathcal{R}(\theta_a(q))^\dagger H(q) \mathcal{R}(\theta_a(q)). \tag{18}$$

Because $\mathcal{R}$ is orthogonal and hence unitary,

$$\lambda_n[A_a(q)] = \lambda_n[H(q)], \qquad n = 1, 2, \tag{19}$$

mathematically in exact arithmetic. The finite calculation nevertheless retains the maximum observed sorted-band defect as an implementation residual.

The mean energy $m(q)$ is invariant under conjugation. Defining $Y_a(q) = A_a(q) - m(q)I_2$, one has $Y_a = \mathcal{R}^\dagger X \mathcal{R}$ and $\|Y_a(q)\|_F = \|X(q)\|_F$ pointwise. The transformation therefore redistributes matrix orientation but does not alter the local eigenvalue splitting.

2.7 Flat analytic control versus embedded dispersive control

The flat analytic control is a separate retained-space model

$$H_{\text{flat}}(q) = \bar{E} I_2 + D, \qquad D = D^T \in \mathbb{R}^{2\times 2}, \quad \operatorname{tr} D = 0, \quad D \neq 0. \tag{20}$$

It has no parent dispersion and no non-onsite reference hopping. Under the sinusoidal gauge attack, its continuum restricted correlation has a Bessel-function expression derived in Section 3.4. The embedded control of Eqs. (7)–(14) is different: its retained field is dispersive, its frame comes from an ambient spectral subspace, and its restricted correlation has no assumed J_0 identity.

These layers must not be conflated. The flat calculation is an analytic control; the ρ family is a parent-model robustness channel; and mesh refinement is a numerical discretization channel.

3 Numerical method and mathematical observables

3.1 Overview of the calculation map

For one request $(N, \rho, \{a_r\}, \text{gates}, \text{tolerances})$, the production calculation performs the composition

$$K_\rho(q_j) \longrightarrow (\lambda_n(q_j), V_j) \longrightarrow F_j \longrightarrow H_j \longrightarrow A_{a_r,j} \longrightarrow \begin{cases} \text{restricted-alignment observables,} \\ \text{cyclic Fourier powers,} \\ \text{ordinary truncation consequences.} \end{cases} \tag{21}$$

The parent is assembled explicitly, diagonalized by a Hermitian eigensolver, and then qualified before any gauge-amplitude point is evaluated. Parent preparation is reused within an amplitude study so that changing a cannot silently change the parent, transported frame, or reference Fourier field.

3.2 Closed-loop polar frame transport

Let $V_j \in \mathbb{C}^{4\times 2}$ be a raw orthonormal spectral frame. Neighboring frames are related through overlap matrices

$$M_j = F_{j-1}^{\dagger} V_j \in \mathbb{C}^{2\times 2}. \tag{22}$$

For a singular-value decomposition $M_j = L_j \Sigma_j R_j^{\dagger}$, the unitary polar factor is $Q_j = L_j R_j^{\dagger}$. A right action based on Q_j chooses the member of the same retained subspace that is closest to the preceding transported frame. The minimum singular value over all links diagnoses whether this polar step is numerically resolved. The installed request uses a rank-relative tolerance to decide numerical rank and a separate validation-relative tolerance for backend residual checks.

Transport around the periodic loop generally leaves an endpoint mismatch $W \in U(2)$. With a Schur or spectral representation

$$W = Z \operatorname{diag}(e^{i\varphi_1}, e^{i\varphi_2}) Z^{\dagger}, \qquad -\pi < \varphi_n \le \pi, \tag{23}$$

principal-branch phases are distributed along the loop. At mesh fraction j/N, the correction has the form

$$C_j = Z \operatorname{diag}\left(e^{-ij\varphi_1/N}, e^{-ij\varphi_2/N}\right) Z^{\dagger}. \tag{24}$$

The resulting frame is periodic under the declared identity endpoint sewing. This is a frame construction, not a change of spectral subspace.

Qualification retains, rather than hides, the following diagnostics:

$$\sigma_{\min} = \min_{j,n} \sigma_n(M_j), \tag{25}$$

$$d_{\text{unitary}} = \max_j \left\| Q_j^{\dagger} Q_j - I_2 \right\|_{\max}, \tag{26}$$

$$d_{\text{closure}} = \left\| \text{computed closure} - \text{declared closure} \right\|_{\max}, \tag{27}$$

$$d_F = \max_j \left\| F_j^{\dagger} F_j - I_2 \right\|_F, \tag{28}$$

$$d_H = \max_j \left\| H_j - H_j^{\dagger} \right\|_{\max}. \tag{29}$$

A backend exception, numerical rank failure, policy threshold failure, external-gap failure, and successful qualification are five different outcomes. They are not encoded as a single boolean.

3.3 Restricted global alignment

The comparison does not admit the local gauge $\mathcal{R}(\theta_a(q))$ as a recovery map. Instead it admits one global real rotation

$$G(\phi) = \mathcal{R}(\phi), \qquad \phi \in [0, \pi), \tag{30}$$

where the interval has length π because conjugation of a traceless rank-two matrix has a doubled-angle periodicity.

For discrete traceless fields X_j and $Y_{a,j}$, define

$$c_{N,a}(\phi) = \frac{1}{2N} \sum_{j=0}^{N-1} \Re \operatorname{tr}\left[X_j^{\dagger} G(\phi)^{\dagger} Y_{a,j} G(\phi) \right]. \tag{31}$$

Rank-two real rotations imply an exact second-harmonic representation

$$c_{N,a}(\phi) = c_0 + c_c \cos(2\phi) + c_s \sin(2\phi). \tag{32}$$

The coefficients are formed directly from the matrix fields, and a continuous harmonic maximizer returns

$$c_{N,a}^{\max} = \max_{\phi} c_{N,a}(\phi), \qquad \phi_{N,a}^{\star} \in \arg\max_{\phi} c_{N,a}(\phi). \tag{33}$$

The maximizing argument can be absent when the harmonic is degenerate under the declared absolute and relative degeneracy tolerances. An absent argument is a scientific outcome, not the number zero.

The normalized discrete maximum correlation is

$$\mu_{N,a} = \frac{c_{N,a}^{\max}}{\kappa_N}, \qquad -1 \le \mu_{N,a} \le 1, \tag{34}$$

up to bounded binary64 roundoff. Values just outside the interval by no more than $32\epsilon_{64}$ are clamped; a larger excursion is a floating-point failure. The identity amplitude $a = 0$ is installed as the exact invariant $\mu_{N,0} = 1$ in the production route.

3.4 Two non-equivalent restricted thresholds

The first comparison allows a nonnegative scalar s together with the global rotation. Its normalized mean squared loss is

$$\mathcal{L}_N(s, \phi; a) = \frac{1}{2N} \sum_j \left\| Y_{a,j} - sG(\phi) X_j G(\phi)^{\dagger} \right\|_F^2 . \tag{35}$$

Because X and Y_a have the same total traceless power κ_N, minimizing the quadratic in $s \ge 0$ yields

$$s_{N,a}^{\star} = \max(0, \mu_{N,a}), \qquad \mathcal{L}_{N,\min}(a) = \kappa_N \left[1 - \max(0, \mu_{N,a})^2\right] . \tag{36}$$

The retained operator threshold is therefore

$$\tau_{\mathrm{op},N}(a) = \sqrt{\mathcal{L}_{N,\min}(a)}. \tag{37}$$

The second comparison fixes the traceless scale, as required by exact spectrum preservation, and minimizes only over the global rotation. Its threshold is

$$\tau_{\mathrm{joint},N}(a) = \sqrt{2\kappa_N[1 - \mu_{N,a}]}. \tag{38}$$

Equations (37) and (38) answer different questions. The first asks for the best globally rotated and rescaled traceless operator; the second asks for the best globally rotated operator at the exact spectral scale. The report never replaces one by the other.

The production route explicitly reconstructs the matrix loss at $(s_{N,a}^{\star}, \phi_{N,a}^{\star})$ and retains

$$d_{\mathrm{recon}}(a) = \left| \mathcal{L}_N(s_{N,a}^{\star}, \phi_{N,a}^{\star}; a) - \mathcal{L}_{N,\min}(a) \right| \tag{39}$$

as a residual channel.

3.5 Flat-parent analytic formulas

For the flat traceless control D, the gauge rotation changes its orientation through twice the angle $\theta_a(q)$. Averaging the aligned correlation over the continuous Brillouin zone gives

$$\frac{1}{1} \int_{-1/2}^{1/2} \cos[2a \sin(2\pi q)] \, dq = J_0(2a), \tag{40}$$

where J_0 is the Bessel function of the first kind. Consequently,

$$\mu^{\mathrm{flat}}_{\infty,a} = J_0(2a), \tag{41}$$

$$\tau^{\mathrm{flat}}_{\mathrm{op},\infty}(a) = \sqrt{\kappa\,[1 - J_0(2a)^2]}, \tag{42}$$

$$\tau^{\mathrm{flat}}_{\mathrm{joint},\infty}(a) = \sqrt{2\kappa\,[1 - J_0(2a)]}. \tag{43}$$

These formulas are analytic comparators only for the flat control. The dispersive embedded parent does not inherit them.

3.6 Finite cyclic Fourier representation

For a sampled matrix field $B_j = B(q_j)$ on Eq. (4), the declared discrete hopping blocks are

$$B_R = \frac{1}{N}\sum_{j=0}^{N-1} e^{-2\pi i q_j R} B_j, \qquad B_j = \sum_{R\in\mathcal{R}_N} e^{2\pi i q_j R} B_R, \tag{44}$$

with a centered cyclic representative inventory $\mathcal{R}_N$. The normalized block power is

$$p_R[B] = \frac{1}{2}\,\|B_R\|_F^2\,, \tag{45}$$

and the non-onsite weight is

$$W_{\neq 0}[B] = \sum_{R\in\mathcal{R}_N\setminus\{0\}} p_R[B]. \tag{46}$$

The reference and comparison weights are retained separately. Their difference is not called an error unless a particular comparison explicitly defines one.

The Parseval residual is computed from the same normalization:

$$d_{\mathrm{Parseval}}[B] = \left| \frac{1}{2N}\sum_j \|B_j\|_F^2 - \sum_{R\in\mathcal{R}_N} p_R[B] \right|. \tag{47}$$

Small residuals establish consistency of the finite transform and normalization. They do not establish continuum convergence.

For even N, $R = -N/2$ is a centered Nyquist representative that is self-conjugate only modulo N. Treating it as an ordinary integer displacement would impose the wrong Hermitian-pairing semantics. The truncation channel therefore removes this cyclic Nyquist block, retains its power separately, and converts the remaining inventory to an ordinary conjugate-closed model satisfying

$$B_{-R} = B_R^\dagger. \tag{48}$$

3.7 Parent-hopping family

The parent-family inventory is

$$\rho \in \{0, 0.25, 0.5, 0.75, 1\}. \tag{49}$$

For each ρ, the complete procedure—parent diagonalization, gap gate, frame transport, retained projection, gauge-amplitude sweep, restricted alignment, and Fourier analysis—is repeated. It would be incorrect to reuse the $\rho = 1$ frame or projected Hamiltonian at another family member. The family therefore measures sensitivity to the ambient parent-hopping homotopy, not merely a scalar interpolation of final observables.

3.8 Nested momentum refinement

The production momentum ladder is

$$N \in \{32, 64, 128, 256, 512\}, \tag{50}$$

with exact doubling. The fine mesh contains every coarse point. At shared points, the study compares parent eigenvalues and spectral projectors rather than raw eigenvectors, because raw eigenvectors are gauge dependent. The shared-point projector defect is schematically

$$d_P(N, 2N) = \max_{q \in \mathcal{Q}_N} \|P_N(q) - P_{2N}(q)\|_F \,. \tag{51}$$

For each adjacent pair $(N, 2N)$ the retained step contains separate maxima for parent qualification diagnostics, shared parent eigenvalues, shared projectors, correlation, operator thresholds, hopping weights, individual hopping powers, maximizing angles, band defects, loss-reconstruction defects, and Parseval residuals. Angle changes are compared only when the maximizing-angle dispositions are compatible; degeneracy status is retained separately.

Let $d_r(N, 2N)$ denote any declared change metric and η_r its threshold. A step passes only if every change metric and every fine-mesh residual satisfies its own bound:

$$\mathrm{pass}(N, 2N) = \bigwedge_r [d_r(N, 2N) \le \eta_r]. \tag{52}$$

Convergence requires the declared number of final adjacent pairs to pass. Earlier passing pairs cannot compensate for a failing final pair, and one small observable cannot compensate for another large one.

3.9 Adaptive amplitude refinement

The amplitude domain is $[0, \pi/4]$. Starting from interpolation nodes $\{a_i\}$, every interval $[a_i, a_{i+1}]$ is first tested at its midpoint $m_i = (a_i + a_{i+1})/2$. For a scalar observable f, the midpoint linear-interpolation defect is

$$d_f^{(1/2)}(i) = \left| f(m_i) - \frac{f(a_i) + f(a_{i+1})}{2} \right|. \tag{53}$$

The same construction is applied separately to correlation, both thresholds, comparison non-onsite weight, and every comparison hopping power. For the hopping-power channel, the maximum is taken over the full displacement inventory after verifying that the inventories agree.

Only failing intervals are refined. If all midpoint checks pass, the node inventory is frozen and validation moves to disjoint quarter points

$$v_{i,1} = a_i + \frac{1}{4}(a_{i+1} - a_i), \qquad v_{i,3} = a_i + \frac{3}{4}(a_{i+1} - a_i). \tag{54}$$

For $\xi \in \{1/4, 3/4\}$ the validation defect is

$$d_f^{(\xi)}(i) = \left| f(a_i + \xi \Delta a_i) - [(1 - \xi) f(a_i) + \xi f(a_{i+1})] \right|. \tag{55}$$

Convergence requires both a passing midpoint level and a passing disjoint quarter-point validation on that same frozen node inventory. A midpoint pass alone is insufficient. The closed alternatives are qualification stop, converged, refinement-level exhaustion, and evaluated-amplitude-cap exhaustion. No alternative is silently relabeled as convergence.

3.10 Ordinary-lattice truncation consequences

After Nyquist removal, let the ordinary hopping model be

$$B^{\mathrm{ord}}(q) = \sum_{R\in\mathcal{R}_{\mathrm{ord}}} e^{2\pi iqR} B_R, \qquad B_{-R} = B_R^\dagger. \tag{56}$$

For an integer cutoff R_c, the symmetric truncation is

$$B^{(R_c)}(q) = \sum_{|R|\le R_c} e^{2\pi iqR} B_R, \tag{57}$$

with ordinary omitted power

$$P^{\mathrm{ord}}_{>R_c}[B] = \sum_{|R|>R_c} \frac{1}{2} \|B_R\|_F^2 . \tag{58}$$

The total omitted-power diagnostic adds the separately retained Nyquist conversion power,

$$P^{\mathrm{total}}_{>R_c}[B] = P^{\mathrm{ord}}_{>R_c}[B] + P_{\mathrm{Nyq}}[B]. \tag{59}$$

This addition is a bookkeeping identity for the finite conversion; it is not an ordinary-lattice hopping coefficient at $R = -N/2$.

Truncation is evaluated on a shifted mesh of size $M = 1025$,

$$\widetilde{q}_\ell = -\frac{1}{2} + \frac{\ell+\omega}{M}, \qquad \omega = \frac{1}{N+1}, \qquad \ell = 0,\dots,M-1, \tag{60}$$

which is checked to be disjoint from the construction mesh. At those points the study retains four norms for each relevant difference: maximum normalized Frobenius error, root-mean-square normalized Frobenius error, maximum sorted-band error, and root-mean-square sorted-band error. For matrix fields C_ℓ and D_ℓ,

$$E_{H,\max}(C,D) = \max_\ell \frac{\|C_\ell - D_\ell\|_F}{\sqrt{2}}, \tag{61}$$

$$E_{H,\mathrm{rms}}(C,D) = \sqrt{\frac{1}{2M}\sum_\ell \|C_\ell - D_\ell\|_F^2}, \tag{62}$$

$$E_{\lambda,\max}(C,D) = \max_{\ell,n} |\lambda_n(C_\ell) - \lambda_n(D_\ell)| , \tag{63}$$

$$E_{\lambda,\mathrm{rms}}(C,D) = \sqrt{\frac{1}{M}\sum_\ell \sum_{n=1}^{2} [\lambda_n(C_\ell) - \lambda_n(D_\ell)]^2}. \tag{64}$$

The implementation checks the finite-dimensional inequality that the maximum sorted-band error does not exceed the maximum Frobenius error beyond a binary64 roundoff allowance.

Three matrix comparisons remain distinct. The interpolation channel compares the full finite comparison Fourier interpolant with a direct gauge action on the interpolated reference. The truncation channel compares truncated and complete comparison interpolants. The total channel compares the truncated comparison with the direct gauge reference. Their errors may be numerically similar in a well-resolved case, but they are not the same defined quantity.

3.11 Canonical aggregate and lifecycle

The calculation retains the parent family, momentum refinement, adaptive amplitude outcome, and truncation outcome in one immutable typed aggregate. Channel variants are closed: no type registry, null padding, or untyped dictionary substitutes for a missing scientific outcome. Canonical serialization follows

$$\texttt{_mapping} \longrightarrow \texttt{to_dict} \longrightarrow \texttt{to_json}, \tag{65}$$

with the inverse path

$$\texttt{_from_mapping} \longrightarrow \texttt{from_dict} \longrightarrow \texttt{from_json}. \tag{66}$$

The calculation script checks byte-for-byte canonical round trip before atomic publication. Retention owns and authenticates those bytes and the canonical visualization projection. The strictly downstream renderer admits stored verification and does not rerun scientific verification.

4 Independent verification and admission

4.1 Independence requirement

The verifier is intentionally more than a schema checker. It reconstructs expected values through direct finite formulas and compares them with the canonical calculation record. The independent route may use NumPy primitives and the declared mathematical model, but it must not call the production parent constructor, frame-transport owner, analysis owner, Fourier owner, request-mesh owner, parent-family owner, adaptive-refinement owner, interpolation owner, cyclic-to-ordinary conversion owner, truncation owner, or gauge-action owner. Reusing those owners would test serialization around the same implementation rather than independently test the implementation.

The verification tolerance is an installed policy of

$$\varepsilon_{\mathrm{verify}} = 10^{-10}. \tag{67}$$

It is not read from retained bytes and cannot be enlarged by an execution artifact. Scientific study thresholds, such as 1.01×10^{-8} for refinement, have a different meaning and do not replace Eq. (67).

4.2 Direct parent and spectral reference

The reference route evaluates Eq. (7) directly at every declared q_j and uses an independent call to Hermitian diagonalization. It reconstructs

$$\Delta_{\mathrm{ext},N}, \quad \{\lambda_n(q_j)\}_{n=1}^{4}, \quad P_j = V_j V_j^{\dagger}, \tag{68}$$

and independently applies the external-gap disposition. The comparison uses ordered eigenvalues and projectors where gauge invariance is required; it never demands equality of arbitrary raw eigenvector phases.

4.3 Direct Gram–polar transport reference

For each neighboring overlap matrix, the reference computes singular values and the unitary polar factor directly. It carries the frame around the loop, diagonalizes the closure mismatch, applies the declared principal-phase branch, and distributes closure phases using Eq. (24). It then reconstructs the transported frame, retained matrices, trace/traceless split, splitting scale, and every retained transport diagnostic.

This route distinguishes:

- external-gap failure before transport;
- numerical rank failure indicated by the transport status;
- backend failure with operation, reason, and link index;
- policy threshold failure when a numerically resolved minimum singular value is not greater than the installed bound; and
- a fully qualified parent with all scalar and phase diagnostics.

The observed and reference outcomes must have the same exact variant type before variant-specific numerical fields are compared.

4.4 Direct alignment and Fourier reference

The verifier reconstructs the gauge rotation from elementary sine and cosine formulas, forms A_a by direct conjugation, and evaluates pointwise eigenvalues. It derives the second-harmonic coefficients in Eq. (32) from residual matrices, solves the continuous harmonic maximum analytically, reconstructs both thresholds, and directly evaluates the matrix loss at the maximizing rotation.

Fourier coefficients are reconstructed from Eq. (44) by explicit finite sums. Reference and comparison block powers, displacement inventories, non-onsite weights, and Parseval residuals are compared separately. Agreement of total weight cannot hide a misplaced individual hopping block; both inventory equality and maximum individual power defect are checked.

For every amplitude point, the verifier compares at least:

maximum sorted-band defect, κ_N, $\mu_{N,a}$, $\tau_{\mathrm{op},N}(a)$, $\tau_{\mathrm{joint},N}(a)$,
$d_{\mathrm{recon}}(a)$, $W_{\neq 0}[H]$, $W_{\neq 0}[A_a]$, $d_{\mathrm{Parseval}}[H]$, $d_{\mathrm{Parseval}}[A_a]$,
harmonic coefficients, maximum, period, status, and maximizing argument,
every reference and comparison displacement power.

The identity of a harmonic outcome and its numerical coefficients are separate checks.

4.5 Family verification

The family verifier first compares the exact ordered ρ inventory. It then executes a complete independent analysis for each family member. A passing member at one ρ does not compensate for a failure at another. The family-level visualization projection is generated independently from the reference family, and the vector and matrix portions of that projection are compared under Eq. (67).

The canonical family bytes are separately hashed even though they are nested inside the aggregate. This nested identity makes the family boundary explicit and prevents a renderer from accepting a family projection that was not derived from the authenticated family record.

4.6 Momentum-refinement verification

The reference route rebuilds every mesh analysis independently and re-evaluates every adjacent-mesh metric. It checks exact mesh doubling, preservation of every non-mesh control, complete coverage of the declared mesh inventory, and equality of the retained step disposition with Eq. (52). It also independently determines whether the required final number of stable refinements has passed and checks that the closed result variant is correspondingly *converged*, *unresolved*, or *qualification stopped*.

This distinction matters because the refinement result can be numerically unresolved even while band-equivalence, Fourier, and Parseval residuals are near binary64 roundoff. A separate parent-diagnostic change may still exceed its declared threshold.

4.7 Adaptive-amplitude verification

The adaptive verifier reconstructs the exact midpoint and quarter-point inventories from the retained nodes, independently evaluates every direct point, and recomputes every linear-interpolation defect. It checks that only failed midpoint intervals or failed quarter points were promoted into the next node inventory. It also checks the evaluation cap before adding new amplitudes.

A converged variant is admitted only if the final midpoint level passes and a disjoint quarter-point validation on the same node inventory also passes. If the level cap or sample cap is reached first, the corresponding exhaustion variant must be retained. This prevents a resource stop from being misreported as convergence.

4.8 Truncation verification

The independent truncation route evaluates finite Fourier sums directly, identifies the centered even-mesh Nyquist representative, removes it from the ordinary inventory, checks ordinary conjugate closure, and recomputes its separate power. It constructs the shifted evaluation mesh, verifies nonoverlap with the construction mesh, performs direct finite interpolation, applies each symmetric cutoff, and recomputes every norm in Eq. (64).

Reference truncation, comparison truncation, interpolation, and total comparison errors are verified as different channels. The verifier also checks the closed truncation variant: qualification stop cannot carry completed ranges, while completion must carry the entire amplitude and cutoff inventories.

4.9 Canonical bytes and stored-verification admission

Let b_C be canonical calculation bytes, b_F canonical nested family bytes, and b_V canonical aggregate visualization-data bytes. The retention package owns these canonical records, the visualization projection, and its authentication. Verification retains

$$h_C = \mathrm{SHA256}(b_C), \qquad h_F = \mathrm{SHA256}(b_F), \qquad h_V = \mathrm{SHA256}(b_V). \tag{69}$$

On readback, the admission path decodes each typed record, re-encodes it canonically, and requires exact byte equality before checking its digest. Therefore a semantically similar but noncanonical JSON document is not silently normalized into an authenticated artifact.

Stored admission then requires:

1. exact canonical calculation bytes;
2. exact canonical verification bytes;
3. matching calculation and visualization digests;
4. the closed expected component inventory; and
5. a passing aggregate verification.

Only after those conditions hold may rendering proceed. The local visualization package is strictly downstream of retention: it consumes the admitted projection and stored verification result, does not rerun the verifier, and does not reinterpret a failed component.

4.10 Interpretation of failure

A failed independent comparison is evidence of a route disagreement under the installed policy. It does not by itself identify whether the producer, the independent reference, the comparison convention, or the installed tolerance is at fault. Diagnosis must name the exact observation and preserve the failing artifact. The first candidate execution failed precisely in this sense, as detailed in Section 5.5. No tolerance was widened and no failed observation was discarded. After exact-identity remediation, a distinct no-overwrite successor passed all four aggregate components; that later pass does not rewrite or invalidate the diagnostic history of the first attempt.

5 Verified unsealed calculation, numerical observations, and disposition

5.1 Status statement

The values and figures in this section come from the authorized unsealed successor execution `direct-embedded-rank2-controlled-v2`. They are calculated finite numerical values, not placeholders and not literature values. Its canonical calculation bytes are identical to those preserved by the failed `v1` attempt. The `v2` aggregate independently verified at the unchanged installed tolerance, and the renderer admitted the stored verification before producing the six PNG artifacts. Accordingly, the plots are authenticated projections of bounded finite software results; they are not sealed evidence, material validation, continuum certificates, or publication acceptance.

5.2 Parent-family qualification and endpoint observations

Every sampled family member passed the production external-gap and transport gates. At the largest gauge amplitude $a = \pi/4$, the candidate values were:

ρ	$\Delta_{\mathrm{ext},256}$	$\sigma_{\min}$	$\mu_{256,\pi/4}$	τ_{op}	$W_{\neq 0}[A_{\pi/4}]$
0.00	1.343658302	1.000000000	0.472001216	0.378394511	0.143182406
0.25	1.270016236	0.999999941	0.474619562	0.379835494	0.145249698
0.50	1.197626387	0.999999754	0.482141898	0.384103485	0.151446266
0.75	1.126501977	0.999999423	0.493670477	0.391045366	0.161756230
1.00	1.056646121	0.999998929	0.507990809	0.400439426	0.176153250

The external gap decreases across this finite ρ inventory but remains above the declared gate $\delta_{\mathrm{ext}} = 1$. The minimum transport singular value also decreases while remaining well above 0.99. At $a = \pi/4$, the restricted correlation increases with ρ in this candidate, while both the operator threshold and comparison non-onsite hopping weight also increase. These are simultaneous observed trends, not a proof of monotonicity between sampled ρ values and not a statement of causal identity between the threshold and hopping weight.

The following three figures are the authenticated parent-family projections retained by the closed visualization inventory.

5.3 Momentum refinement was unresolved

The momentum result has the exact closed outcome variant `MomentumRefinementUnresolved`. The four adjacent mesh steps retained the following principal maxima:

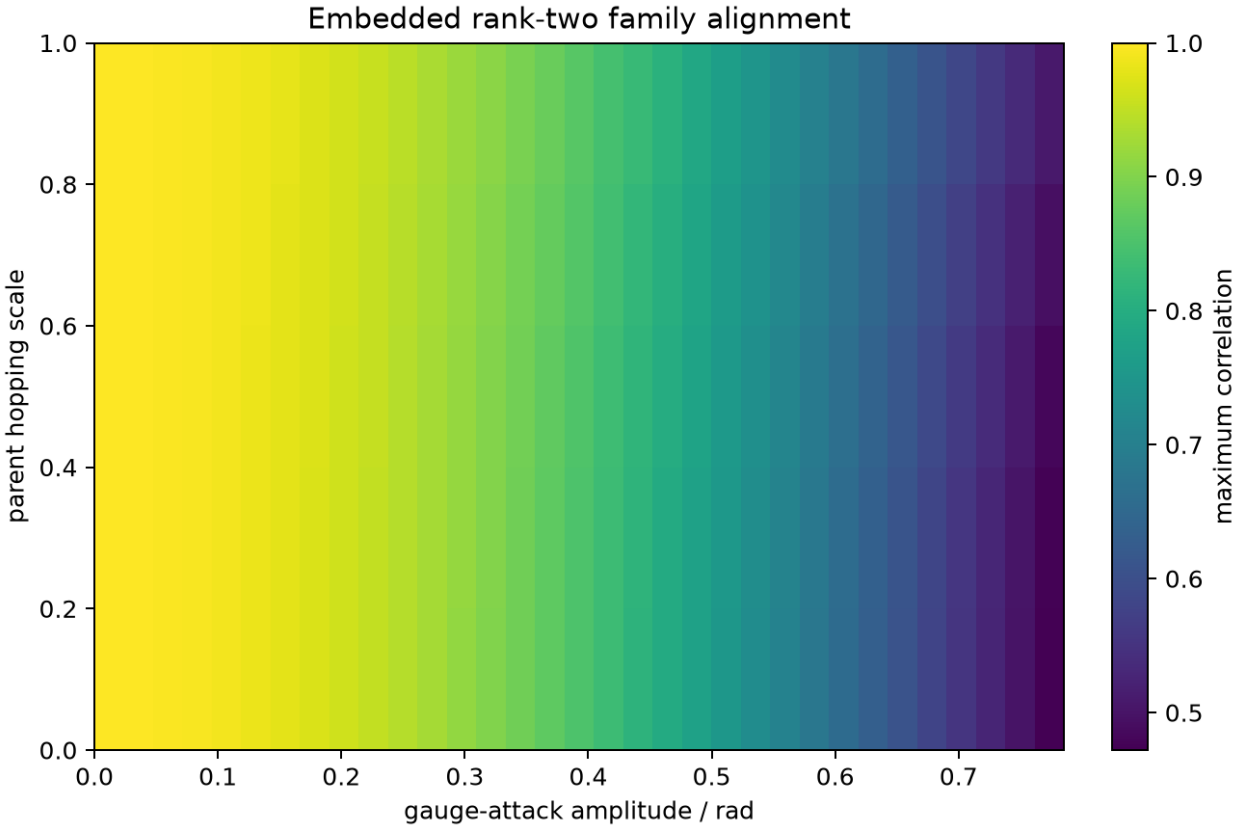


Figure 1: Authenticated parent-family alignment landscape. It displays the finite amplitude trajectories of restricted correlation and the two threshold definitions across the five sampled parent-hopping scales.

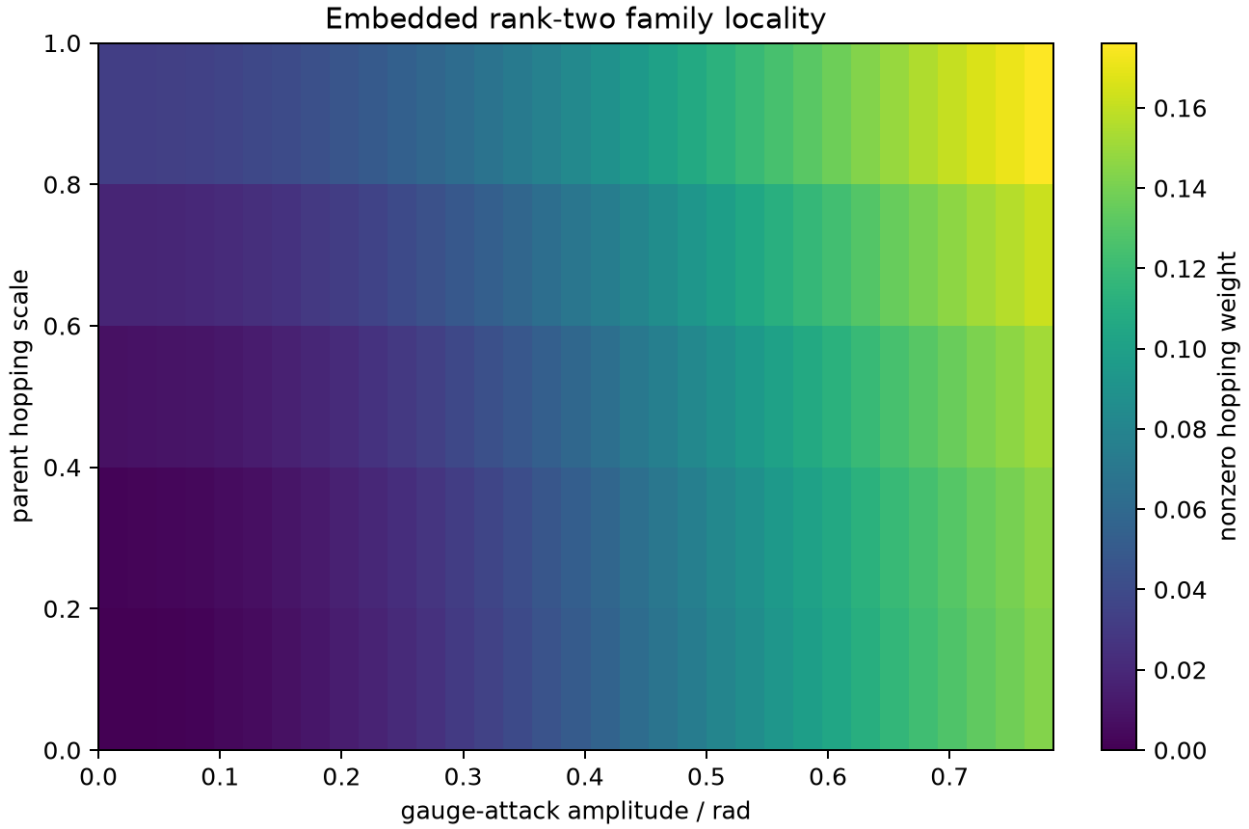


Figure 2: Authenticated parent-family locality landscape. Reference and comparison non-onsite hopping weights remain distinct from operator thresholds even when their sampled trends are visually correlated.

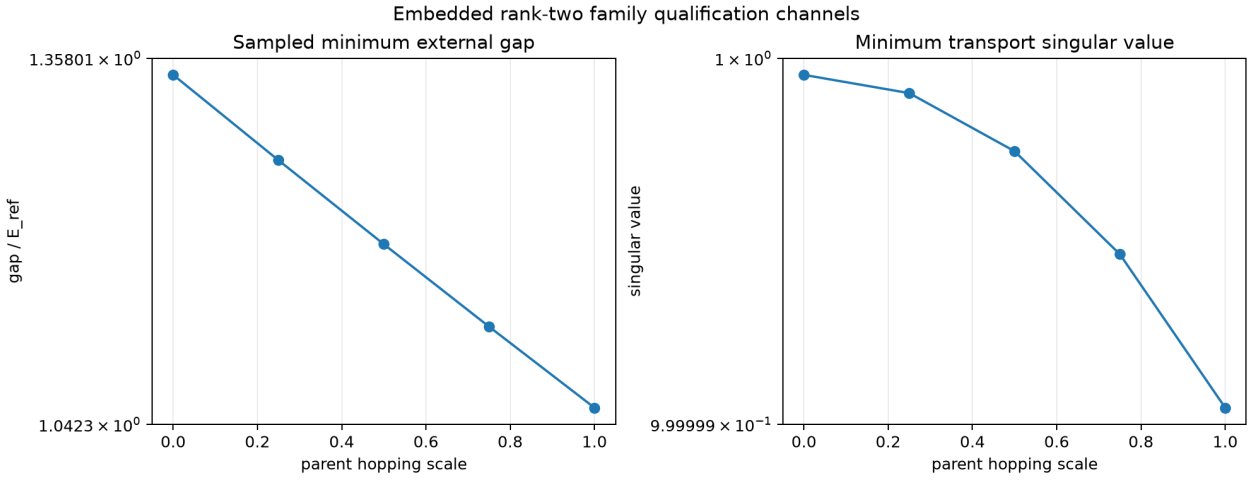


Figure 3: Authenticated parent-family qualification diagnostics. External-gap and transport controls are shown for the finite sampled ρ inventory; they do not certify the continuous domain between samples.

$N \to 2N$	parent diagnostic	correlation	threshold	hopping weight	max residual
$32 \to 64$	7.2983×10^{-3}	6.7028×10^{-11}	3.1413×10^{-11}	6.8701×10^{-10}	1.11×10^{-15}
$64 \to 128$	3.6586×10^{-3}	1.6883×10^{-11}	7.9118×10^{-12}	1.7225×10^{-10}	9.99×10^{-16}
$128 \to 256$	1.8305×10^{-3}	4.2286×10^{-12}	1.9821×10^{-12}	4.3095×10^{-11}	1.11×10^{-15}
$256 \to 512$	9.1540×10^{-4}	1.0576×10^{-12}	4.9544×10^{-13}	1.0776×10^{-11}	1.33×10^{-15}

Here "max residual" summarizes the largest displayed band-equivalence, loss-reconstruction, or Parseval-scale residual to order of magnitude; the typed record retains those channels separately. Shared parent eigenvalue and projector defects at nested shared points were exactly zero in the retained binary64 values, and maximizing angle changes were at most 2.22×10^{-16} after the first step.

Despite the small correlation, threshold, hopping, band, reconstruction, and Parseval changes, every step failed the common draft threshold 1.01×10^{-8} because the parent-diagnostic change remained much larger. The last two steps therefore did not supply the required two final passing refinements. Calling this channel converged would be false. The sequence of parent-diagnostic changes is approximately halved under mesh doubling, which is useful diagnostic information but not a convergence pass under the declared absolute criterion.

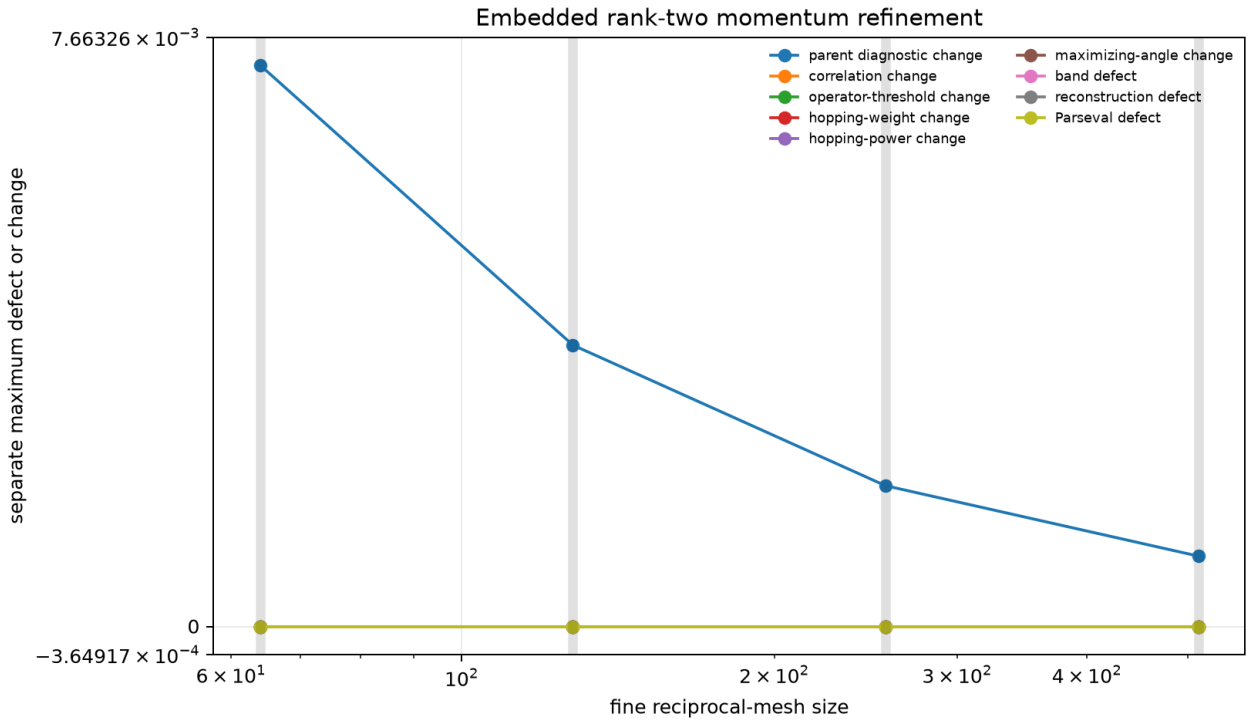


Figure 4: Authenticated momentum-refinement diagnostics. Independent verification of the retained values does not change the scientific outcome `MomentumRefinementUnresolved`.

5.4 Adaptive amplitude refinement exhausted its level cap

The adaptive result has the exact closed outcome variant `AdaptiveAmplitudeLevelsExhausted`. No disjoint quarter-point validation was reached because no midpoint level passed. The retained level maxima were:

level	nodes	d_{μ}	$d_{\tau_{\mathrm{op}}}$	$d_{\tau_{\mathrm{joint}}}$	$d_{W_{\neq 0}}$	d_{p_R}
0	9	2.2268×10^{-3}	8.3265×10^{-4}	3.7493×10^{-4}	8.8021×10^{-4}	8.8021×10^{-4}
1	17	5.5841×10^{-4}	2.0947×10^{-4}	9.6263×10^{-5}	2.2216×10^{-4}	2.2216×10^{-4}
2	33	1.3971×10^{-4}	5.2475×10^{-5}	2.4376×10^{-5}	5.5671×10^{-5}	5.5671×10^{-5}
3	65	3.4934×10^{-5}	1.3129×10^{-5}	6.1323×10^{-6}	1.3926×10^{-5}	1.3926×10^{-5}

Each level directly evaluated the midpoint-augmented inventory, ending with 129 unique amplitudes at level three. Every interval failed at every level under the 1.01×10^{-8} per-channel thresholds. The roughly factor-of-four reduction in linear interpolation defects after interval halving is consistent with smooth second-order interpolation behavior, but it remains far from the declared pass criterion. Since midpoint acceptance was never achieved, the absence of quarter-point validation is correctly rendered as "no disjoint validation retained," not as a zero defect or passing check.

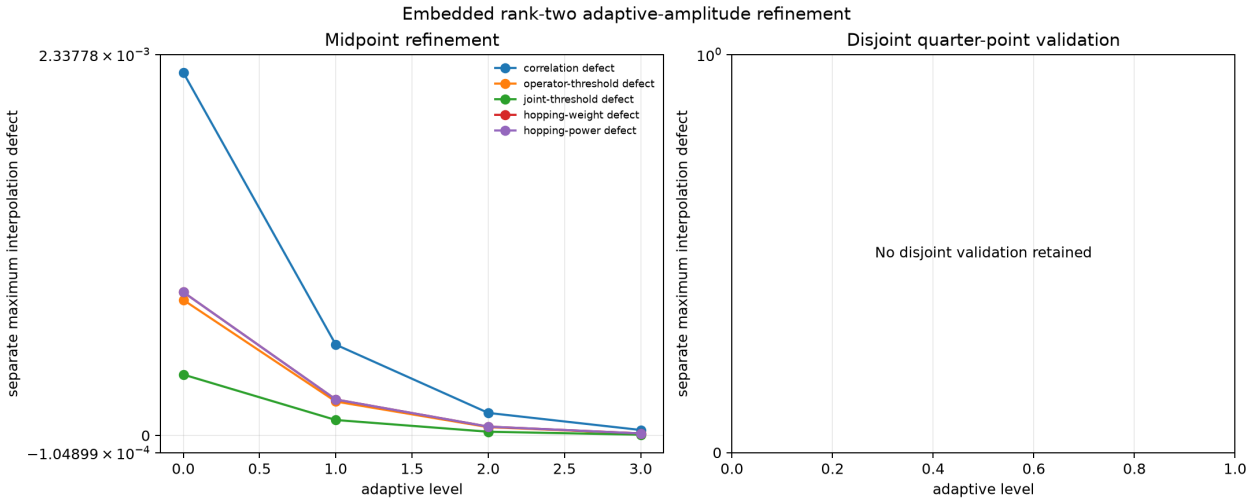


Figure 5: Authenticated adaptive-amplitude diagnostics. Midpoint and disjoint validation channels remain logically separate, and level exhaustion is not convergence.

5.5 Truncation channel completed

The truncation result has the exact closed outcome variant `TruncationCompleted`. Its shifted evaluation mesh used

$$M = 1025, \qquad \omega = \frac{1}{257} = 0.00389105058\ldots, \tag{70}$$

and the minimum periodic construction–evaluation separation was 1.48287×10^{-8}. The small separation is positive, so the meshes are disjoint in binary64; it should not be interpreted as a generous separation margin.

At $a = 0$, comparison interpolation error is exactly zero in the retained candidate, as expected for the identity gauge. At $a = \pi/4$, the maximum normalized Hamiltonian interpolation error is 8.59×10^{-16} and the maximum sorted-band interpolation error is 1.33×10^{-15}. The comparison Nyquist power is 2.16×10^{-33} at that endpoint, while the reference Nyquist power is 1.61×10^{-33}.

For the full parent and $a = \pi/4$, the cutoff-zero and cutoff-sixteen endpoint values are:

R_c	reference omitted power	comparison omitted power	$E^{\mathrm{total}}_{H,\max}$	$E^{\mathrm{total}}_{\lambda,\max}$
0	0.0323839143	0.1761532501	0.5590843271	0.5249763398
16	3.15×10^{-31}	3.74×10^{-31}	1.84×10^{-15}	2.66×10^{-15}

Thus the candidate calculation exhibits a large representation-dependent inflation of non-onsite power at $a = \pi/4$, while a sufficiently long retained range reconstructs the finite data to near binary64 precision. This statement is restricted to the candidate finite model and does not imply a continuum or material result.

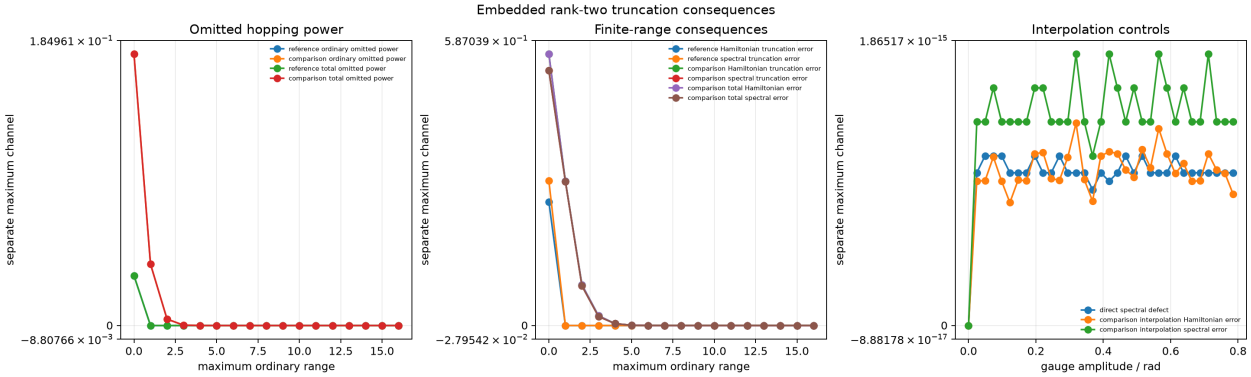


Figure 6: Authenticated truncation consequences on the disjoint shifted mesh. Matrix, spectral, interpolation, and omitted-power channels remain separately identified.

5.6 Verification history and exact-identity remediation

The `v1` aggregate verification failed only in the parent-family component; the momentum, adaptive-amplitude, and truncation verification components passed their installed comparisons. Within the family, members at $\rho = 0.5$ and $\rho = 0.75$ failed the point scalar comparison at the exact identity amplitude $a = 0$:

$$d_{\rho=0.5} = 6.5330815607 \times 10^{-9}, \tag{71}$$

$$d_{\rho=0.75} = 6.7004386123 \times 10^{-9}, \tag{72}$$

against the installed tolerance

$$\varepsilon_{\text{verify}} = 10^{-10}. \tag{73}$$

The family visualization matrix comparison inherited the larger $6.7004386123 \times 10^{-9}$ defect.

Diagnosis localized the disagreement to the two threshold scalars at $a = 0$. The production route enforces the exact mathematical identity $\mu_{N,0} = 1$ and therefore retains

$$\tau_{\text{op},N}(0) = \tau_{\text{joint},N}(0) = 0. \tag{74}$$

For those two family members, the independent route retained a correlation infinitesimally below one before applying square roots in Eqs. (37) and (38). A perturbation of order machine epsilon in $1 - \mu$ becomes a threshold of order $\sqrt{\epsilon_{64}}$, producing the observed 10^{-9} scale. This was the diagnosed numerical mechanism in `v1`. No tolerance was enlarged, no failing bytes were rewritten, and rendering remained blocked for that execution.

The independent reference was then corrected to encode the exact $a = 0$ identity as an independently justified mathematical invariant, and a regression test was added for $\rho = 0.5$ and $\rho = 0.75$. The distinct no-overwrite `v2` execution retained calculation SHA-256

`29c1a2bdd7ec85b721cb9c9248ba0961cd1472278cf79a33f641eaa9e657dd01`,

which is byte-identical to the `v1` calculation record. All four aggregate verification components passed under $\varepsilon_{\text{verify}} = 10^{-10}$; the stored verification SHA-256 is

`cf74d9fd35f3419bec4bd98ac5535396337f37df9fcc8de56a9ae5e369a92691`.

Thus the remediation changed independent handling of a known exact identity, not the calculated scientific values or the installed tolerance. The failed `v1` artifacts remain distinguishable and preserved.

5.7 Overall technical outcome

The verified unsealed calculation supplies informative finite observations: all parent family members qualified; the momentum study was unresolved because of a slowly changing parent diagnostic; adaptive amplitude refinement exhausted its level cap; and the truncation channel completed. The lifecycle outcome is therefore **bounded independent verification passed; stored rendering admitted; momentum and amplitude convergence not established**. This is more precise than either "the run failed" or "the science passed."

6 Limitations, assumptions, and nonclaims

6.1 Synthetic model limitation

The parent Hamiltonian is synthetic. Its entries use an arbitrary positive energy scale, and no orbital labels, fitted parameters, crystal chemistry, pseudopotential, exchange–correlation approximation, or experimental uncertainty is attached to them. The model demonstrates representation and locality mechanisms in a controlled finite setting. It does not predict any named semiconductor.

6.2 Sampled rather than continuous qualification

The external gap is minimized on a finite mesh. A passing sampled gap does not prove a positive gap at every $q \in \mathcal{B}$. The momentum ladder provides finite stability information, but its candidate outcome is unresolved and does not certify a continuum limit. Likewise, finite amplitude inventories do not establish all extrema between samples. Adaptive interpolation defects quantify a declared linear-interpolation question only.

6.3 Gauge and comparison restriction

The admitted recovery is one momentum-independent real rotation, optionally accompanied by a nonnegative scalar in the operator-threshold problem. It is not the full group of momentum-dependent $U(2)$ gauge transformations, not a complex global-unitary search, not a disentanglement algorithm, and not maximally localized Wannier gauge optimization. A large restricted residual therefore means that the two matrix fields cannot be well identified by this restricted action; it does not mean their pointwise spectra differ.

Conversely, exact pointwise spectral equivalence does not imply equality of represented matrix fields or Fourier coefficients. Hopping redistribution under a gauge action is a representation effect inside the retained subspace. It is not evidence that the parent physical bands changed.

6.4 Finite Fourier and truncation limitation

The hopping inventory first obtained from N samples is cyclic. Removal of the even-mesh Nyquist representative and conversion to an ordinary conjugate-closed inventory are finite-data operations. A first-passing cutoff, if one were selected, would be conditional on the construction mesh, evaluation mesh, norm, gauge, amplitude, tolerance, and maximum tested range. It would not prove compact support or an asymptotic decay law.

Omitted block power, matrix interpolation error, truncation error, and sorted-band error are distinct. None should be substituted for another without a proved relationship. The finite Frobenius bound checked by the implementation is a consistency constraint, not a model-adequacy theorem.

6.5 Floating-point limitation

All retained numerics are binary64. Hermitian diagonalization, singular values, polar factors, Schur phases, Fourier summation, norms, and square roots are subject to rounding. The first candidate verification failure illustrates a particularly important numerical feature: an $O(\epsilon_{64})$ perturbation in a nonnegative squared quantity can become an $O(\sqrt{\epsilon_{64}})$ perturbation after a square root. Exact invariants such as the identity gauge must therefore be encoded deliberately rather than inferred from a nearly exact floating-point cancellation.

6.6 No uncertainty or statistical inference

The study has no stochastic sampling, parameter posterior, confidence interval, uncertainty propagation, or model discrepancy distribution. Its thresholds are declared numerical policies, not statistical significance levels. A smooth trend across five ρ values is not a regression result or a probability statement.

6.7 No formal-execution equivalence claim

Related Lean developments can formalize algebraic identities for finite periodic models. They do not certify NumPy execution, binary64 behavior, package versions, canonical JSON, SHA-256 manifests, or plot generation. Conversely, passing Python tests does not prove the corresponding mathematical theorem. Formal mathematics, numerical verification, and retained evidence remain complementary but distinct layers.

6.8 No evidence promotion

Nothing in this report seals, promotes, publishes, or releases either execution. Both remain unsealed. The failed `v1` verification still prevents rendering from that execution. The authenticated plots in this report come only from the passing stored `v2` verification, but rendering does not itself seal or scientifically validate the calculation. Rebuilding this PDF documents the present repository state; it does not change the evidence status of any calculation artifact.

7 Provenance, lifecycle, and reproduction

7.1 Direct stage sequence

The maintained lifecycle consists of three direct module-scope programs:

1. `run_01_calculation.py` constructs the four-channel aggregate and writes canonical calculation bytes plus execution metadata;

2. `run_02_verification.py` independently reconstructs all four channels, writes canonical verification bytes and a manifest, and requires the aggregate verification to pass; and

3. `run_03_render.py` admits the stored verification and emits the closed six-figure inventory without rerunning scientific verification.

The scripts contain direct composition rather than a generic workflow engine. They do not define command-line parsers, custom JSON encoders, helper classes, or hidden scientific defaults.

7.2 No-overwrite and atomic publication

For each stage, a final path D and incomplete path D_{inc} are declared in advance. The stage refuses to begin if either exists. Outputs are written under D_{inc} and renamed atomically to D only after all stage checks complete. This design prevents an interrupted stage from masquerading as complete and prevents a rerun from silently rewriting retained bytes.

The first candidate exposed a subtle stage-level limitation: verification bytes were published before the script raised on the aggregate passing condition. Those bytes are useful failure diagnostics but cannot be treated as a passing verification directory. They must not be deleted or rewritten merely to make the lifecycle look successful. A remediated calculation must use a distinct execution identity or an explicitly authorized migration.

7.3 Execution artifact inventories

The first unsealed directory,

`executions/direct-embedded-rank2-controlled-v1/`,

preserves the canonical calculation, the failing verification and manifest, execution metadata, and `UNSEALED.md`. It has no `figures/` directory because rendering was not admitted.

The authorized successor directory is

`executions/direct-embedded-rank2-controlled-v2/`.

It contains:

- `execution.json`, identifying the calculation family, execution identity, result path, result digest, script digest, package versions, claim boundary, and unsealed status;
- `results/calculation.json`, the canonical aggregate whose SHA-256 is `29c1a2bdd7ec85b721cb9c9248ba0961cd1472278cf79a33f641eaa9e657dd01`;
- `verification/verification.json`, the passing aggregate verification whose SHA-256 is `cf74d9fd35f3419bec4bd98ac5535396337f37df9fcc8de56a9ae5e369a92691`;
- `verification/manifest.json`, which binds the stage paths and digests;
- `figures/rendering.json`, which binds the closed figure inventory to the stored verification; six authenticated PNG files; and
- `UNSEALED.md`, which explicitly denies sealed-evidence status.

The v1 and v2 calculation JSON files are byte-identical. Their verification artifacts differ because v2 incorporates the exact zero-amplitude identity remediation. The current three direct scripts byte-match the producer digests recorded by v2. The preserved v1 metadata instead record historical calculation- and verification-script SHA-256 values c19178d2cbfdc8af094f7628deef1b5c8237383a64ec572a2d301ad25ba1533c and 58e2a9fcf8fea3b07252d91509ebcf2a6eb8e97c3fb8973698f8e7844b22ee01. Those values necessarily differ from the successor scripts after the execution identity and independent exact-zero route changed. The current checkout retains their digests, not separate byte copies of the two historical v1 producer scripts. This is a provenance limitation of the failed attempt; the immutable v1 directory is not rewritten to disguise it.

7.4 Canonical identity checks

For any retained typed record x, the stage requires

$$\mathrm{encode}(\mathrm{decode}(\mathrm{encode}(x))) = \mathrm{encode}(x) \tag{75}$$

byte for byte. The manifest then records SHA-256 of those exact canonical bytes. On admission, both canonical equality and digest equality are checked. These steps prevent an equivalent-but-differently-encoded JSON representation from acquiring the identity of the retained artifact.

7.5 Software environment

The execution metadata records installed versions of ksdft2effmass, NumPy, Project Koios Optimization, Project Koios PhysKit, and SciPy. The script digest binds the exact direct calculation entry point. The current report does not claim that package version strings or script hashes alone capture the complete machine environment, BLAS/LAPACK implementation, compiler options, operating system, or hardware. A sealed reproduction package would require a separately reviewed environment and provenance policy.

7.6 Reproduction commands and authorization

From the repository root, the intended commands are

```
uv run --project python python \
  calculations/periodic1d/rank2/\
  band-equivalence-versus-hopping-locality/run_01_calculation.py
uv run --project python python \
  calculations/periodic1d/rank2/\
  band-equivalence-versus-hopping-locality/run_02_verification.py
uv run --project python python \
  calculations/periodic1d/rank2/\
  band-equivalence-versus-hopping-locality/run_03_render.py
```

These commands are documentary, not blanket authorization. A new production execution requires explicit human authorization and a new no-overwrite execution identity. The existing v1 and v2 identities cannot be reused while their directories exist.

7.7 Report build

The technical report is built from this directory with a warnings-visible LaTeX toolchain such as

```
latexmk -pdf -interaction=nonstopmode -halt-on-error technical-report.tex
```

The source checks for authenticated `v2` PNG paths. Missing files produce explicit boxes rather than silently substituting development figures. Building the PDF reads neither the calculation JSON nor the verification JSON and does not alter their status.

7.8 Separation from historical donors

Historical workflow modules and development figures may explain design history, but they are not inputs to the direct renderer and are not authenticated by this aggregate. Development plots under `analysis/development/figures/` must not be copied into the authenticated slots. Migration crosswalks should identify their provenance and eventual retirement disposition without pretending that historical outputs were produced by the installed direct lifecycle.

A Symbol glossary and installed control ledger

A.1 Principal mathematical symbols

Symbol	Meaning
q	Dimensionless reduced crystal momentum in $[-1/2, 1/2)$, periodic with period one.
k	Physical inverse-length wavevector, related to q by Eq. (3).
a_0, G	Physical lattice period and reciprocal-lattice period $G = 2\pi/a_0$.
j, N	Momentum-sample index and construction-mesh size.
R	Integer ordinary cell displacement or a representative of a finite cyclic displacement class, as stated.
ρ	Homotopy parameter and parent non-onsite hopping scale in $[0, 1]$.
$\mathcal{H}$	A continuous homotopy map satisfying the endpoint conditions in Eq. (6).
a	Dimensionless gauge-attack amplitude in $[0, \pi/4]$.
$K_\rho(q)$	Hermitian 4×4 ambient parent Hamiltonian; as a joint function of (q, ρ), it is the installed parent-hopping homotopy.
$\lambda_n(q)$	Ordered parent or retained eigenvalue, with the represented matrix stated by context.
V_j	Raw 4×2 retained eigenvector frame.
F_j	Closed-loop transported periodic 4×2 retained frame.
P_j	Rank-two spectral projector $F_j F_j^\dagger$.
H_j	Represented retained Hamiltonian $F_j^\dagger K_j F_j$.
m_j, X_j	Half trace of H_j and its traceless part.
κ_N	Mean normalized squared Frobenius power of the traceless reference field.
$\theta_a(q)$	Sinusoidal gauge angle $a \sin(2\pi q)$.
$\mathcal{R}(\theta)$	Real rotation in $SO(2)$; calligraphic type distinguishes it from displacement R.
$A_a(q)$, $Y_a(q)$	Gauge-conjugated comparison Hamiltonian and its traceless part.
$G(\phi)$	One momentum-independent real rotation admitted by the restricted comparison.
$c_{N,a}(\phi)$	Normalized matrix inner-product correlation before maximizing over ϕ.
$\mu_{N,a}$	Maximum restricted correlation divided by κ_N.

Symbol	Meaning
τ_{op}	Minimum operator threshold allowing a nonnegative global scale.
τ_{joint}	Exact-spectrum joint threshold with traceless scale fixed.
B_R	Matrix-valued Fourier hopping block at displacement R.
$p_R[B]$	Normalized block power $\|B_R\|_F^2/2$.
$W_{\neq 0}[B]$	Sum of all non-onsite cyclic block powers.
R_c	Symmetric ordinary-lattice truncation cutoff.
$E_{H,*}$	Normalized Frobenius represented-Hamiltonian error.
$E_{\lambda,*}$	Sorted-band spectral error.
ϵ_{64}	Binary64 machine epsilon.
$\varepsilon_{\mathrm{verify}}$	Fixed independent-comparison tolerance 10^{-10}.

A.2 Installed finite controls

Control	Installed value
Execution identities	`direct-embedded-rank2-controlled-v1` for the preserved failed verification attempt and `direct-embedded-rank2-controlled-v2` for the passing unsealed successor.
Ambient and retained ranks	4 and 2.
Parent family construction mesh	$N = 256$.
Parent-family amplitude inventory	32 equal intervals, hence 33 points on $[0, \pi/4]$.
Parent-hopping scales	$(0, 0.25, 0.5, 0.75, 1)$.
Momentum ladder	$(32, 64, 128, 256, 512)$.
Momentum amplitude inventory	64 equal intervals, hence 65 points on $[0, \pi/4]$.
Required final stable refinements	2.
Adaptive construction mesh	$N = 128$.
Adaptive initial amplitude inventory	8 equal intervals, hence 9 nodes.
Maximum adaptive levels	4.
Maximum unique evaluated amplitudes	129.
Truncation cutoff inventory	Every integer $R_c = 0, 1, \ldots, 16$.
Truncation evaluation mesh	$M = 1025$ shifted points.
External-gap gate	$\delta_{\mathrm{ext}} = 1.0$ parent-energy unit.
Minimum transport singular-value gate	Strictly greater than 0.99.
Frame absolute tolerance	10^{-10}.
Transport rank-relative tolerance	10^{-12}.
Transport validation-relative tolerance	10^{-10}.
Harmonic absolute degeneracy tolerance	10^{-14}.
Harmonic relative degeneracy tolerance	10^{-13}.

Control	Installed value
Momentum and amplitude study thresholds	1.01×10^{-8} independently for every declared channel.
Independent verification tolerance	10^{-10}, fixed in installed code rather than retained request bytes.
Endpoint sewing	Identity in the fixed ambient basis.
Closure phase branch	Principal phase branch.
Energy convention	Arbitrary positive synthetic parent-energy unit; no physical-unit calibration.
Reciprocal convention	Centered, half-open, period-one reduced coordinate.

A.3 Array shapes and finite inventories

At construction size N, the parent matrix field has shape $(N, 4, 4)$, ordered parent eigenvalues have shape $(N, 4)$, raw and transported frames have shape $(N, 4, 2)$, and retained matrix fields have shape $(N, 2, 2)$. These shapes are part of the operational contract. Arrays retained by domain objects are copied to canonical numerical dtypes and made read-only.

For one qualified amplitude point, the scalar record contains the amplitude, maximum band defect, splitting scale, harmonic maximum, normalized restricted correlation, two thresholds, loss-reconstruction defect, reference and comparison non-onsite weights, reference and comparison Parseval defects, and the paired reference/comparison hopping powers over the complete cyclic displacement inventory.

For one completed truncation amplitude point, the record additionally contains reference and comparison Nyquist powers, ordinary Hermiticity defects, direct spectral-equivalence defect on the shifted mesh, four interpolation errors, and one range record for every cutoff. Each range record retains reference and comparison omitted powers and separate reference truncation, comparison truncation, and total comparison matrix and spectral errors.

A.4 Closed scientific outcomes

Parent qualification. Exactly one of external-gap failure, transport-backend failure, transport numerically unresolved, transport-threshold failure, or qualified parent is retained. The variants carry different fields because the available scientific information differs.

Momentum refinement. Exactly one of qualification stopped, converged, or unresolved is retained. Convergence requires the declared final stable-refinement count; exhaustion is not encoded as a successful result.

Adaptive amplitude refinement. Exactly one of qualification stopped, converged, levels exhausted, or evaluations exhausted is retained. Convergence requires both midpoint acceptance and disjoint quarter-point validation.

Truncation. Exactly one of qualification stopped or completed is retained. A stopped result cannot carry fabricated cutoff consequences.

Aggregate verification. The component inventory is exactly parent family, momentum refinement, adaptive amplitude, and truncation, plus the prescribed nested visualization-projection checks. A missing, reordered, substituted, or additional component is rejected.

A.5 Error-channel crosswalk

Channel	Mathematical object	What it does not establish
External-gap qualification	Sampled $\min(\lambda_3 - \lambda_2)$	Continuous isolation between samples.
Transport singular value	Minimum singular value of neighbor overlaps	Momentum convergence of all observables.
Band defect	Sorted eigenvalue difference under gauge conjugation	Matrix-field equality or locality equality.
Restricted correlation	Global-$SO(2)$ matrix correlation	Optimality over momentum-dependent $U(2)$.
Operator threshold	Scaled restricted matrix residual	Exact-spectrum matrix residual.
Joint threshold	Fixed-scale restricted matrix residual	Hopping truncation error.
Parseval defect	Finite Fourier normalization consistency	Continuum Fourier convergence.
Non-onsite weight	Sum of finite cyclic powers for $R \neq 0$	A unique ordinary localization length.
Momentum change	Difference between adjacent finite meshes	Parent-model uncertainty.
Amplitude defect	Linear interpolation defect in a	Global analytic regularity or all unsampled extrema.
Nyquist power	Power removed during cyclic-to-ordinary conversion	Ordinary hopping at literal displacement $-N/2$.
Ordinary omitted power	Sum of blocks beyond R_c	Total represented-Hamiltonian error without interpolation/conversion context.
Matrix truncation error	Frobenius difference of represented fields	Sorted-band error equality.
Spectral truncation error	Difference of sorted retained eigenvalues	Equality of represented matrices.
SHA-256 digest	Identity of canonical bytes	Numerical correctness or scientific validity.
Independent verification	Agreement of two finite software routes	Material validity, publication acceptance, or evidence sealing.